Jiarui Mao
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EDUCATION

University of North Carolina at Chapel Hill September 2022 - May 2026
B.S. in Biostatistics, Minor in Neuroscience
GPA: 3.83

RELEVANT COURSEWORK

Practical Machine Learning for Biosignal Analysis ● Introduction to Biostatistics ● Linear Algebra for Applications ● Advanced Calculus ● Differential Equations ● Probability and Statistical Inference ● Advanced Neuropharmacology

RESEARCH EXPERIENCE

YAP LAB, Department of Radiology **May 2024 - Present**
Research assistant | University of North Carolina at Chapel Hill
Principle Investigator: Pew-thian Yap, PhD.
Project: Development of structure-function coupling in early infant stage
Conducted large-scale computational modeling and network analysis to investigate multimodal neuroimaging data (fMRI, dMRI, EEG) for over 300+ subjects and 1000+ scans
Developed and optimized analysis pipelines for high-dimensional datasets (~20k+ vertices) on High-Performance Computing (HPC) cluster, utilizing parallel computing for large-scale data processing and model fitting.
Designed and created all data visualizations and figures for publication, resulting in a first-author manuscript in preparation for submission to a peer-reviewed neuroscience journal.
Project: Optimal shrinkage denoising in high-résolution fMRI
Utilized AFNI, FSL, and MATLAB for fMRI image processing, gradient construction, and quantitative evaluation of spatial-temporal signal preservation.
Benchmarked pipelines to compare the efficiency and accuracy of multiple denoising algorithms (e.g., MPPCA, OSSVD, NORDIC) across large-scale datasets.
Designed visualizations and posters to communicate results, which were presented as abstracts at international conferences.

Hopfinger Lab, Department of Neuroscience **January 2023 - Present**
Research assistant | University of North Carolina at Chapel Hill
Principle Investigator: Joe Hopfinger, PhD.
Project: ERP Evidence of Attentional Suppression of Unpredictable Distractors
Design and program cognitive attention paradigms in PsychoPy and OpenSesame, implementing randomized distractor tasks for participants to investigate predictive control mechanisms.
Collect, monitor, and preprocess high-quality neural recordings from over 40 participants, contributing ~10 hours per week to experimental setup, data acquisition, and figure preparation.
Preprocess and analyze EEG data using R studio and MatLab, presenting results in weekly lab meetings.

Pasteur Institute of Shanghai, Department of Immunology **May 2021 - Aug 2022**
Research assistant | University of Chinese Academy of Sciences, Shanghai
Principle Investigator: Haikun Wang, PhD.
Project: Regulatory Role of RIPK1 in T-Cell homeostasis
Project summary:
Performed intravenous injections, tissue collection, and immunostaining in rodent models over a 6-month experimental period to study RIPK1-mediated immune regulation.
Utilized Western blotting, gel electrophoresis, and flow cytometry to quantify protein expression and characterize T-cell subpopulations and immunoglobulin levels across biological samples.
Process data and analyze for immune response using data-processing software (SAS, SPSS, R)

PRESENTATIONS & ABSTRACTS

Jiarui M. Khoi H. Kim-Han T. Sahar A. Pew-Thian Y., *Early Structure-Function Coupling and Its Association with Myelination and E-I Balance*, SFN, 2025
Jiarui M. Khoi H. Kim-Han T. Patrick T. Guoye L. Sahar A. Pew-Thian Y. *Precision Functional Gradients via Optimal Shrinkage Denoising,* ISMRM, 2025

Julie L. Khoi H. Kim-Han T. **Jiarui M.** Patrick T. Guoye L. Sahar A. Pew-Thian Y. *Structural Gradients of the Developing Brain Connectome*, ISMRM, 2025

Jiarui M. Khoi H.& Pew-Thian Y., *Lowering thermal noise in functional MRI with optimal shrinkage denoising*, Radiology Departmental Research Symposium, 2024

Jiarui M. Oleh D., Jenny Q., & Rachel P., *2-Br-LSD Treatment of Mood and Depressive Disorders,* Annual Celebration of Undergraduate Research (ACUR), 2024

AWARDS

Office of Undergraduate Research(OUR) Travel Award 2025

Awarded a **$300 travel scholarship** by the **UNC Office of Undergraduate Research** to support presentation of undergraduate research at a national conference.

Kimball King Undergraduate Research Award 2025

Awarded a $500 scholarship to support senior honor thesis project.

International Society for Magnetic Resonance in Medicine(ISMRM) Travel Award 2025

Awarded a $500 scholarship and membership to support presentation of research at the annual meeting.

Society for Neuroscience(SfN) Member 2025

Sponsored by Dr. Pew-Thian Yap's Lab, Department of Radiology, University of North Carolina at Chapel Hill.

Dean's List 2022 - 2025

Top 20% of undergraduate student at University of North Carolina at Chapel Hill (minimum 3.50 GPA)

LEADERSHIP&SERVICE

International Neuroscience and Psychology Scholars Club (INPSC) March 2023 – August 2025

Founder and President | Chapel Hill

Summary:

Facilitated mentorship pair-ups between graduate students and undergraduates for over 30+ club members; helps non-native English speakers to adapt in UNC

Organized and moderated faculty speaker panels, facilitating discussions on research design, graduate training, and academic career development for 50+ attendees.

Angelcare Breast Cancer Volunteer Group May 2023 – August 2024

Writer, Editor | Shanghai

Summary:

Authored and edited 30+ science communication articles on breast cancer awareness and patient care for the NGO's online platform, reaching 10,000+ views.

Assisted with major community events and conducted weekly interviews with 20+ patients.

UNC Learning center March 2023 - December 2023

Academic coach | Chapel Hill

Summary:

Delivered 50 + tutoring sessions to students about math, neuroscience and physics, guided students in effective test preparation techniques and study strategies;

Serve as an Undergraduate Learning Assistant (ULA) for NSCI273 hosting weekly office hours. Design in class-worksheet, practice computer programs, final exam and midterm questions for over 60+ students to use.